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Mathematical Language in Finance: Pricing European Options Using Stochastic Calculus

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Index

<i>Introduction</i>	3
<i>1. Two Fundamental concepts: Absence of arbitrage and Completeness</i>	6
1.1 Arbitrage free market: the standing assumption	6
1.2 European Options	7
1.3 Mathematical formulation of the No Arbitrage Condition	7
1.4 Market Completeness	9
1.5 Kenneth Arrow and State-Contingent Claims	9
1.6 The Mathematical Shift: European Call Options	10
1.7 Linear Algebra Version and Proof of $D \text{ span } \mathbb{R}^M$	11
<i>2. From Assumptions to Models</i>	13
2.1 Mathematical Prerequisites	13
2.2 Assumptions	17
2.3 European Call Options and Contingent claims	19
2.4 Pricing the claim before Maturity	20
2.5 Black-Scholes Model market and absence of arbitrage	20
2.6 Black-Scholes Pricing Formula	22
2.7 Solution of the Black-Scholes Equation	26
2.8 Limits of the Black-Scholes model	29
2.9 Implications of the limits and possible extensions	33
<i>3. Black-Scholes in Python</i>	35
3.1 Bloomberg: the inputs	35
3.2 Data Extraction	36
3.3 Python Implementation	37
<i>Conclusion</i>	39
<i>Bibliography</i>	40

Introduction

In 1602, the first modern financial market was established in Amsterdam allowing public companies to trade shares and bonds. Since then, people have tried to guess the behavior of the market, wondering what lies behind the movement of an asset price. What could drive its going up and down, and they have found answers. The value of a stock can be influenced by a myriad of factors: geopolitical situations, the companies' performance, the trends in different markets, the legal rules of a state and so on. There are millions of self-influencing variables and this can make predicting an asset's trajectory feel like an impossible task.

Addressing this complexity, Benjamin Graham proposed a possible answer writing in 1949 "The intelligent Investor"¹. This masterpiece was famously claimed by the great Warren Buffet to be: "by far, the best book on investing ever written"². In the eighth chapter, the writer presents an allegory that could be of help for understanding a crucial point in finance. Graham says: imagine you own a stake in a private worth \$1000 and you have a partner named Mr. Market. Every day, he makes assumptions about the value of your stocks and he offers whether to buy your interest or sell you his. It often happens that Mr. Market gives advices based on his swings of optimism and pessimism. The secret for an intelligent investor should be to exploit the extreme fluctuations in behavior of the partner wisely rather than being governed by them. The investor should buy from Mr. Market when he proposes an awfully low price and to sell when he does the opposite. Here, Graham intuitively anticipates something that is called arbitrage, but we will explain in deep this concept later. As it is widely known, the aim of Graham's work was to provide a practical guide for investors to value investing. He advocated for a disciplined focus on intrinsic value, urging investors to make rational decisions rather than emotional ones. His purpose wasn't to present the market performance over a period time and to explain the scheme behind it but we can still take a lesson by his words in that sense. He has written in "The intelligent Investor": "Every Investor must come to terms with one simple fact: The stock market will fluctuate – both up and down – and there's nothing you can do about it."

Indeed, it is not entirely true that nothing can be done. The fact that everything is random doesn't mean that the random cannot be predicted. Nassim Nicholas Taleb once said: "Randomness is not the

¹ Graham B. (2020), The Intelligent Investor, Harper Business (HarperCollins), New York

² Graham B. (2020), The Intelligent Investor, Harper Business (HarperCollins), New York, cover endorsement by Warren Buffet

absence of pattern, but the presence of a pattern not yet discovered”³. In other words, there is a way to properly model prices that fluctuate randomly. In fact, the behavior of financial products can be studied, analyzed and predicted. Through the years, there have been many theories, this study explores how we can model these seemingly random price fluctuations using stochastic processes in continuous time.

Let’s take a step back in order to understand clearly the relationship behind Finance and Mathematics. Mathematics has long been considered the “language of science” as it has played an essential role in the evolution of knowledge and technology throughout time and space. Compared to this milestone, finance is a relatively young field and it hasn’t yet been fully understood. Many people still view the art of investing as a game. They gamble that tomorrow the price will go up or eventually down and money will be lost. This is why, despite markets have a big potential for intelligent investors, they are often reduced as a way of firing off strategies (at this point just casually). This work adopts a different prospective: finance can be modelled rigorously through mathematical structures, moving beyond common stereotypes. The late 19th century marked a pivotal turning point as the fields of finance and mathematics began to find common ground. For that purpose, it will be recalled the first volume of the journal “Finance and Stochastics” that Springer-Verlag has been publishing since 1997⁴:

*“Nearly a century ago, Louis Bachelier published his thesis *Théorie de la Spéculation*, Ann. Sci. École Norm. Sup. 3 (1900) [1], in which he invented Brownian motion as a tool for the analysis of financial markets. A.N. Kolmogorov, in his own landmark work “*Über die analytischen Methoden in der Wahrscheinlichkeitsrechnung*”, Math. Annalen 104 (1931), pp.415-458, credits Bachelier with the first systematic study of stochastic processes in continuous time. But in addition, Bachelier’s thesis marks the beginning of the theory of option pricing, now an integral part of modern finance. Thus the year 1900 may be considered as birth date of both Finance and Stochastics. For the first seven decades following Bachelier, Finance and Stochastics followed more or less independently. The theory of stochastic processes grew fast and incorporating classical calculus became a powerful mathematical tool - called stochastic calculus. Finance lay dormant until the middle of the twentieth century, and then was resurrected as an offshoot of general equilibrium theory in economics. With the work in the late 1960s and early 1970s of Black, Merton, Samuelson and Scholes, modelling stock prices as geometric Brownian motion and using this model to study equilibrium and arbitrage pricing, the two disciplines were reunited. Soon it was discovered how well suited stochastic calculus*

³ Taleb, N. N. (2001), *Fooled by randomness: The hidden role of chance in life and in the markets*, Random House, New York

⁴ Finance and Stochastics (1997), Springer-Verlag, Berlin

with its rich mathematical structure - martingale theory, Itô calculus, stochastic integration and PDE's - was for a rigorous analysis of contemporary finance, which would lead one to believe (erroneously) that also these tools were invented with the application to finance in mind. Since then the interplay of these two disciplines has become an ever growing research field with great impact both on the theory and practice of financial markets."

There were parallel findings during the same time-period which were born to be helpful of one another; that's how the field of mathematical finance was eventually developed. The dichotomy of finance and mathematics brought to the creation of a discipline that explain in a modern way many financial behaviors.

1. Two fundamental concepts:

Absence of arbitrage and Completeness

In order to examine the issue in depth, I will follow an approach that emphasizes a central linking element, which illustrates how mathematics serves as an appropriate tool for explaining financial phenomena. However, this should not be taken to imply that such a justification exhausts the subject matter or settles the question entirely.

1.1 Arbitrage free market: the standing assumption

Financial markets operate under specific consistency constraints, and the most fundamental among them is the absence of arbitrage. This principle can be translated into an economic rule, best summarized by Nobel laureate Milton Friedman: "*There is no such thing as a free lunch* ⁵."

In substance, nothing can be obtained without a cost, especially when it appears so. This is not merely a philosophical idea; on the contrary, it is directly applicable to market dynamics. In markets for goods, services and labor, scarcity and trade-offs force choices. Similarly, in financial markets, an investor cannot gain a profit without bearing some cost or risk. No agent can construct a strategy that yields a riskless profit with probability one and zero initial cost. A stock will always be associated with a probability of gains, never a guarantee. Applied to finance, this principle takes on a more precise meaning: *arbitrage* refers specifically to the practice of exploiting price discrepancies across two or more markets to obtain a riskless profit. It is important to emphasize that the absence of arbitrage assumption does not mean that arbitrage opportunities never arise in practice; rather, it means that the model excludes them as a structural condition. In financial theory, the absence of arbitrage is imposed as a fundamental modeling assumption, not an empirical truth. In fact, there are traders whose job is precisely to find "holes in the market" — that is, to identify these arbitrage opportunities and profit from them before the market corrects itself.

⁵ Friedman M. (1975), *There's No Such Thing as a Free Lunch*, Open Court Publishing Company, La Salle (IL)

1.2 European Options

We will introduce stochastic processes in continuous time, in order to price European call options. Since a complete knowledge of these financial instruments is not required to follow the mathematical framework, we will limit ourselves to the essential concepts. Consider a practical scenario: Company A signs a contract with Company B, in which Company B agrees to deliver 1,000 mobile phones to Company A in six months. Company A faces currency risk: it knows the price of the order today, but cannot know the exchange rate six months from now. To hedge against this uncertainty, Company A could enter into a second contract that provides protection against an unfavorable rate while still allowing it to benefit should the market move in its favor. This is precisely what an option is: a financial agreement that grants the buyer the right, but not the obligation, to buy or sell a particular asset at a predetermined price — known as the strike price — within a specified period. Among the various types, a European option can be exercised only on its expiration date, which makes it the most mathematically tractable and the one we will focus on throughout this work. The central objective of our discussion will be to determine how to price such financial products, given that their value is not directly observable on the market.

1.3 Mathematical formulation of the No Arbitrage Condition and Market Model

Now, we'll get to the heart of the matter: the absence of arbitrage. To keep things simple, we will work within a one-period model, since it is sufficient to highlight essential ideas while remaining accessible, and the results extend naturally to much more general models. First, it is necessary to delineate a model. We consider a financial market consisting of N different assets (any of them, for example bonds, stocks, options). As already explained, the market exists at only two points in time, $t=0$ and $t=1$ making the system discrete. The price per unit of asset i at time t will be denoted by $S^i(t)$. We, therefore, have a price vector process $S(t=0,1)$ where the price vector is represented as a column vector:

$$S(t) = \begin{bmatrix} S^1(t) \\ \vdots \\ S^N(t) \end{bmatrix}$$

The randomness in the system is modeled by assuming that there is a finite sample space $\Omega = (\omega_1, \dots, \omega_M)$ and that the probabilities $p_j = P(\omega_j) > 0$ for $j=1, \dots, M$ all strictly positive.

We assume that the price vector S_0 is deterministic and known at time $t=0$. In contrast, the same price vector at $t=1$ depends upon the outcome $\omega \in \Omega$ and $S^i(1)(\omega_j)$ denotes the price per unit of asset i at $t=1$ given ω_j has occurred. This allows us to define the payoff matrix D :

$$D = \begin{bmatrix} S^1(1)(\omega_1) & S^1(1)(\omega_2) & \cdots & S^1(1)(\omega_M) \\ S^2(1)(\omega_1) & S^2(1)(\omega_2) & \cdots & S^2(1)(\omega_M) \\ \vdots & \vdots & \ddots & \vdots \\ S^N(1)(\omega_1) & S^N(1)(\omega_2) & \cdots & S^N(1)(\omega_M) \end{bmatrix}$$

written also as:

$$D = S(1) = [d_1 \quad \cdots \quad d_M]$$

where $d_1, \dots, d_M$ are the columns of D , each representing the prices of all N assets under scenario ω_j . Having defined the model, we now introduce the necessary assumptions to discuss absence of arbitrage. We assume that all asset prices are strictly positive at both dates, is:

$$S^1(0) > 0$$

$$S^i(1)(\omega_j) > 0, \quad j = 1, \dots, M$$

We also introduce the hypothesis of a portfolio as an N dimensional row vector $h = [h^1, \dots, h^N]$ with the interpretation that h^i is the number of units of asset i held. The portfolio is established at $t = 0$ at the deterministic prices S_0 , it is kept and later liquidated at $t = 1$ at the stochastic prices S_1 . The value of the portfolio at time t is given by the stochastic process V defined as:

$$V^h(t) = \sum_{i=1}^N h^i S^i(t) = hS(t), \quad t = 0, 1$$

and evaluated at each scenario ω_i :

$$V^h(t)(\omega_i) = hS(t)(\omega_i) = hd_i = (hD)_i$$

Now we can finally state the definition of an arbitrage portfolio:

The portfolio h is an arbitrage portfolio if it satisfies the conditions:

$$V^h(0) = 0,$$

$$P(V^h(1) \geq 0) = 1,$$

$$P(V^h(1) > 0) > 0.$$

In further detail:

$$V^h(0) = 0,$$

$$V^h(1)(\omega_i) \geq 0, \quad \text{for all } i = 1, \dots, M,$$

$$V^h(1)(\omega_i) > 0, \quad \text{for some } i = 1, \dots, M.$$

By imposing these conditions, we arrive at the formal definition of absence of arbitrage. From what we have achieved, it is clear that this financial conclusion aligns naturally with a mathematical framework: one begins by imposing a fundamental rule – the impossibility of riskless profit– which can be developed in progressively more complex models, driven by stochastic processes.

Our presupposition allows us to translate abstract economic principles into precise mathematical conditions. Such a translation is a primary reason for the adoption of mathematical language in finance; it provides a structured environment in which logical deductions can be made with certainty, even when modeling the inherent uncertainty of the markets.

1.4 Market Completeness

For the sake of simplicity, throughout this thesis, we assume another pillar of asset pricing: market completeness. We begin by outlining the origins of this notion and the questions it addresses. After that, we will applicate it within both financial and mathematical frameworks. At the heart of any market lies a fundamental question; “What can be achieved on the market?”. This question does not reflect the inability of economists to devise strategies for the future, but rather acknowledges the presence of an intrinsic uncertainty that is difficult to incorporate into any model.

We can assume full knowledge of every asset price at $t = 0$, however, it is also true that we cannot know what the future holds at $t = 0$. By $t = 1$, several possible states of the world can manifest, for example there could be a good state where the stock’s price increases slightly as there could be a crisis state or even a boom state, among others. Each scenario results in a different payoff levels and, at the outset, we have no way of knowing which reality will unfold.

1.5 Kenneth Arrow and State-Contingent Claims

In 1953, the theory of complete markets was introduced by the Nobel laureate Kenneth Arrow⁶ who made a radical conceptual move. He argued that in a world of multiple possible future states, the goal should be to smooth consumption across states. To this end, he proposed the idea of a state-contingent commodity, captured by the now-famous example: “*an apple tomorrow if state x occurs*”. This concept of a claim contingent upon the realization of a specific event of the world was a turning point for the modern economic theory. It allows wealth to be allocated separately for every possible state; hence every investor can theoretically insure against all prospective future risks. If tradable claim exists for every possible state of nature, a market is said to be complete. By separating scenarios and creating contingent claims, risk becomes a tradable object and, as a consequence, risk sharing becomes possible. In this way, the economy spans the entire space of uncertainty.

⁶ Arrow K. J. (1953), Le rôle des valeurs boursières pour la répartition la meilleure des risques, Centre National de la Recherche Scientifique, Paris

Let:

M = finite set of states of nature

L = number of physical goods

The commodity space is

$\mathbb{R}^{L \times M}$

An allocation is a vector specifying consumption of each good in each state.

Markets are complete if the price system assigns a price to every element of $\mathbb{R}^{L \times M}$.

This is not merely an economic result as it has a direct, parallel interpretation with financial markets. In this context, Kenneth Arrow's State-contingent commodities correspond to the state-contingent payoffs generated by traded assets. The definition can be restated in purely financial terms as follows:

A market is said complete when it is possible for investors to build a portfolio of existing financial instruments (bonds, shares, options...) that can perfectly replicate any desired future payoff, regardless of the future state of the world.

Equivalently, market completeness requires that every possible payoff vector, specifying how much a claim pays in each future state, can be replicated as a linear combination of the payoffs of the assets available in the market. In mathematical terms, the columns of the payoff matrix D must span $\mathbb{R}^M$, where M is the number of states.

1.6 The Mathematical Shift: European Call Options

We now turn to the mathematical formulation of this discussion and we will see how mathematics allows a clearer and more rigorous articulation of these underlying ideas. In the economic discussion above, we referred to a tradable claim which in financial markets takes the form of a derivative. For us, it will be a European call option defined as follows:

A contingent claim (financial derivative) is any random variable X , defined on Ω

$$\Omega = \{\omega_1, \dots, \omega_M\}$$

$$X: \Omega \rightarrow \mathbb{R}$$

in discrete model, this is equivalent to:

$$X \in \mathbb{R}^M$$

It represents any stochastic quantity of money to be received at $t = 1$. The next step is to discuss how it is possible to generate payment streams at $t=1$ by forming portfolios of the underlying assets. Recall that the underlying assets available in the market are given by the price vectors:

$$S(t) = \begin{bmatrix} S^1(t) \\ \vdots \\ S^N(t) \end{bmatrix}$$

We assume that the market $S^1, \dots, S^N$ is arbitrage free and that there exists a risk-free asset.

Consider a contingent claim X . If there exists a portfolio h , based on the underlying assets, such that:

$$V^h(1) = X, \quad \text{with probability 1.}$$

$$V^h(1)(\omega_j) = X(\omega_j), \quad j = 1, \dots, M.$$

then we say X is replicated, or hedged by h . Such a portfolio h is called a replicating, or hedging portfolio. If every contingent claim can be replicated, we say that the market is complete.

1.7 Linear Algebra Version and Proof of Rows of D span $\mathbb{R}^M$

To fully understand the concept of completeness, it is wise to examine the corresponding result in linear algebra.

The market is complete if and only if the rows of the matrix D span $\mathbb{R}^M$. That is D has rank M .

$$D = \begin{bmatrix} S^1(1)(\omega_1) & S^1(1)(\omega_2) & \cdots & S^1(1)(\omega_M) \\ S^2(1)(\omega_1) & S^2(1)(\omega_2) & \cdots & S^2(1)(\omega_M) \\ \vdots & \vdots & \ddots & \vdots \\ S^N(1)(\omega_1) & S^N(1)(\omega_2) & \cdots & S^N(1)(\omega_M) \end{bmatrix}$$

In other words, the market is complete if and only if the payoff vectors of traded assets are rich enough to generate every possible payoff of any contingent claim X across all states.

For any portfolio h , we view the random variable V^h as a row vector $[V^h(1)(\omega_1), \dots, V^h(1)(\omega_M)]$. As established earlier, at $t=1$ we have:

$$V^h(1) = hD$$

The market is thus complete if and only if, for every random variable X (viewed as a row vector in $\mathbb{R}^M$), the equation:

$$hD = X$$

admits a solution h . Since hD is exactly a linear combination of the rows of D with the components of h as coefficients, the set of all attainable payoffs coincides with the row space of D . Completeness

therefore requires that every vector in R^M can be written as such a linear combination, which is equivalent to saying that the rows of D span R^M , or equivalently that $\text{rank}(D)=M$. This seems like an obvious result but we can try to see the deeper meaning. If there are M possible states of the world, a contingent claim assigns one payoff to each state and can be identified with a vector in R^M . To replicate any such claim, an investor needs a sufficient number of assets whose payoff vectors are linearly independent. If the portfolio does not contain enough independent assets, some payoff vectors cannot be constructed and the market is incomplete. For instance, in a world with $M=3$ states, a portfolio consisting of only $N=2$ linearly independent assets has a row space of dimension at most 2, and therefore cannot span R^3 ; not every payoff vector is attainable. This is precisely what it means, in the language of linear algebra, for the rows of D to fail to span R^M .

2. From the Assumptions to the Models

The following chapter presents a mathematical theorem that is applied daily in financial markets. Through this theorem, we will venture deeper into both the sciences, drawing on advanced mathematics and advanced financial concepts, and we will see how the two connect. The subject of chapter two is the pricing of European call options, financial instruments already introduced in the preceding discussion. To give a preview of what follows: we will first present the main mathematical preliminaries required to arrive at the theorem, then the assumptions and the hypotheses on which it rests and finally the theorem itself along with its resolution. Throughout the chapter, it will become evident, more than once, how powerfully mathematics can describe financial phenomena, even if there are moments where the mathematics had to be carefully adapted to make such a description possible.

2.1 Mathematical Preliminaries

In this section, we introduce the main mathematical objects⁷ in stochastic calculus that will be used to describe pricing dynamics in continuous time. This will give a taste of how mathematics can model financial events. The relevant assumptions will be introduced in parallel as they are needed.

Stochastic process⁸

Before introducing the stochastic differential equations that govern asset price dynamics, we need to make precise what it means for a quantity to evolve randomly over time. The central concept is that of a stochastic process, which provides the mathematical foundation for describing any system whose state is subject to randomness. In this thesis, it is the language in which stock prices, portfolio values and all other time-varying financial quantities will be expressed. Here is the formal definition:

Let $(\Omega, \mathcal{F}, P)$ be a probability space and let $T \subseteq \mathbb{R}$ be an index set (typically $T = [0, T]$ or $T = [0, \infty)$). A stochastic process is a family of random variables $\{X(t); t \in T\}$ where, for each fixed $t \in T$, the map $X(t): \Omega \rightarrow \mathbb{R}$ is $\mathcal{F}$ -measurable.

⁷ To lighten the notation, from now on we drop ω

⁸ The definition of stochastic process, including the underlying probability space $(\Omega, \mathcal{F}, P)$, the index set T , and the notion of sample path, follows the classical presentation in Øksendal (2003, Ch. 2) and Karatzas & Shreve (1991, Ch. 1)

Equivalently, X can be viewed as a function $X: T \times \Omega \rightarrow \mathbb{R}$. For each fixed $\omega \in \Omega$, the map $t \mapsto X(t, \omega)$ is called a *sample path* (or *realisation*) of the process.

Informally, a stochastic process is a collection of random variables indexed by time: at each moment t , the value $X(t)$ ⁹ is drawn from a probability distribution that may depend on the history of the process. When t represents calendar time and $X(t)$ the price of a financial asset, this definition captures precisely the idea that prices evolve randomly yet in a structured, probabilistic way. This concept underpins everything that follows: what will be presented, the stock price $S(t)$, the bond price $B(t)$, the portfolio value $V(t)$, and the derivative price $\Pi(t)$, are all stochastic processes.

Filtration¹⁰

Intuitively, a filtration is simply a mathematical way of encoding the flow of information over time. As time progresses, more is observed about the evolution of a stochastic process, and the set of events that can be distinguished grows accordingly. A filtration captures this accumulation of knowledge in a precisely: at each point in time t , it collects everything that is knowable up to that moment, without any anticipation of future observations. This concept is essential when working with stochastic differential equations because it specifies what information is available at each instant and therefore what a process is allowed to depend on. It follows the mathematical definition:

The symbol $\mathcal{F}_t^X$ denotes the σ -algebra generated by the process X over the interval $[0, t]$, based upon observations of the trajectory $\{X(s); 0 \leq s \leq t\}$: it is possible to decide whether a given event A has occurred or not, then we write this as:

$$A \in \mathcal{F}^X(t)$$

If the value of a given random variable Z can be completely determined given observations of the trajectory $\{X(s); 0 \leq s \leq t\}$, then we can write:

$$Z \in \mathcal{F}^X(t)$$

If Y is a stochastic process such that we have:

$$Y(t) \in \mathcal{F}^X(t) \text{ for all } t \geq 0$$

then we say that Y is adapted to the filtration $\{\mathcal{F}^X(t)\}_{t \geq 0}$

⁹ There exist several notions of convergence in probability. Here we refer to mean square convergence (reference to: Brownian Motion and Stochastic Calculus, Karatzas & Shreve)

¹⁰ The notions of natural filtration $\mathcal{F}^X(t)$, measurable adaptedness of events and random variables, and adapted processes are presented following Øksendal (2003, Ch. 2) and Karatzas & Shreve (1991, Ch. 1). For a thorough measure-theoretic treatment see also Revuz & Yor (1999, Ch. 1).

ODE: Ordinary differential equation¹¹

Let $y: [0, T] \rightarrow \mathbb{R}$ be an unknown function of a single real variable t . An ordinary differential equation of order n is an equation of the form:

$$F(t, y(t), y'(t), y''(t), \dots, y^n(t)) = 0$$

Where F is a given function and n is the order of the equation. A solution is a function $y(t)$ that is n times differentiable on $[0, T]$ and satisfies the equation for all $t \in [0, T]$.

An ODE describes how a quantity changes over time when that change depends only on the current state of the system and on time itself. For example, if the rate of change of a quantity is proportional to its current value, we obtain the simple ODE:

$$dy/dt = \mu y(t)$$

whose solution is $y(t) = y_0 e^{\mu t}$

In mathematical finance, ODEs arise naturally in deterministic models of asset growth. However, since financial systems are subject to randomness, this motivates the generalization to stochastic differential equations.

SDE: Stochastic differential equation¹²

Let $(\Omega, \mathcal{F}, \{\mathcal{F}^X(t)\}_{t \geq 0}, P)$ be a filtered probability space and let $W(t)$ be a standard Brownian motion (we will explain this fundamental concept in the assumptions) adapted to the filtration. A stochastic differential equation (SDE) for an unknown stochastic process $X(t)$ is an equation of the form

$$dX(t) = \mu(t, X(t)) dt + \sigma(t, X(t)) dW(t), \quad X(0) = x_0 \quad (1)$$

where $\mu: [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ is the drift coefficient and $\sigma: [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ is the diffusion coefficient. The equation is understood in the Itô integral sense: it is shorthand for the integral equation:

$$X(t) = x_0 + \int_0^t \mu(s, X(s)) ds + \int_0^t \sigma(s, X(s)) dW(s)$$

where the second integral is an Itô stochastic integral. A solution $X(t)$ is a continuous, adapted stochastic process satisfying this integral equation P -almost surely.

¹¹ The definition of ordinary differential equation of order n , together with the notion of classical solution on an interval $[0, T]$, follows the standard formulation in Coddington & Levinson (1955, Ch. 1) and Teschl (2012, Ch. 1).

¹² The definition of stochastic differential equation in Itô form, including existence and uniqueness conditions for solutions, follows Øksendal (2003, Ch. 5) and Karatzas & Shreve (1991, Ch. 5). See also Protter (2004, Ch. 5) for the semimartingale formulation.

The stochastic differential equation is the central mathematical object of this thesis. It is therefore worth developing a genuine intuition for what it represents. Financial instruments are subject to continuous random shocks – news arrivals, shifts in sentiment, the unpredictable aggregate behavior of market participants. An SDE is the mathematical answer to this reality: it decomposes the infinitesimal evolution of a quantity into a systematic part and a random part. The systematic part is called the drift and it represents the average tendency of the process, the direction it is being pulled in over time. The random part is called the diffusion and it represents the unpredictable fluctuations superimposed on that tendency, driven by a Brownian motion. Neither part alone is sufficient: a model with only drift would be unrealistically deterministic, while a model with only diffusion would have no economic direction whatsoever.

Linear SDE¹³

A Stochastic differential equation is called linear if both the drift and diffusion coefficients are affine functions of the state variable X :

$$dX(t) = (a(t)X + b(t)) dt + (c(t)X(t) + d(t)) dW(t)$$

where $a, b, c, d: [0, T] \rightarrow \mathbb{R}$ are deterministic functions of time. When $b(t) = d(t) = 0$, the equation is called homogeneous linear.

Stochastic integrals¹⁴

Let $(\Omega, F, \{F_t\}_{t \geq 0}, P)$ be a filtered probability space and let $W(t)$ be a standard Brownian motion. Let $\sigma(t, \omega)$ be a stochastic process that is:

1. adapted to the filtration $\{F_t\}$ — meaning $\sigma(t, \cdot)$ depends only on information available up to time t
2. square-integrable — meaning $E \left[\int_0^T \sigma^2(t) dt \right] < \infty$

The Itô stochastic integral of σ with respect to W over $[0, t]$ is defined as the mean-square limit

$$\int_0^t \sigma(s) dW(s) = \lim_{n \rightarrow \infty} \sum_k \sigma(t_k) [W(t_{k+1}) - W(t_k)]$$

where $t_0 = 0 < t_1 < \dots < t_n = t$ is a partition of $[0, t]$ with mesh tending to zero, and the sum is evaluated at the left endpoint t_k of each subinterval. The resulting integral is itself a stochastic process in t .

¹³ The specialisation to linear stochastic differential equations, and the derivation of their explicit solutions via the Itô formula, follows Øksendal (2003, Ch. 5) and Karatzas & Shreve (1991, Ch. 5).

¹⁴ The construction of the stochastic integral with respect to Brownian motion, including the relevant integrability conditions, is presented following Øksendal (2003, Ch. 3) and Protter (2004, Ch. 2). For a more general treatment with respect to semimartingales, see Protter (2004, Ch. 3) and Revuz & Yor (1999, Ch. 4).

PDE: Partial differential equation¹⁵

Let $U \subseteq \mathbb{R}^n \times [0, T]$ be an open domain and let $u: U \rightarrow \mathbb{R}$ be an unknown function of the independent variables $(x_1, \dots, x_n, t)$. A partial differential equation (PDE) of order k is an equation of the form

$$F(x_1, \dots, x_n, t, u, \partial u / \partial x_1, \dots, \partial^k u / \partial x_n^k) = 0$$

A solution is a function $u(x_1, \dots, x_n, t)$ that is k times continuously differentiable on U and satisfies the equation pointwise.

In what follows, we will restrict attention to solutions that are twice continuously differentiable. Unlike ODEs, PDEs generally require both initial conditions and boundary conditions to single out a unique solution. It is essential to mention PDEs as the Black-Scholes model reduces to a second-order parabolic PDE, which fully determines the price of a European option at any prior to expiry.

2.2 Assumptions

Some assumptions are required to define the precise mathematical environment in which the Black-Scholes derivation can be carried out rigorously. Without them, the key steps would either break down or yield non-unique results.

We will assume $\alpha(t)$ and $\sigma(t)$ from the (1) formula are deterministic functions of time. This ensures that drift and volatility are known in advance and do not depend on the state of the market. Under this assumption, the Black-Scholes PDE has deterministic coefficients, making it analytically tractable and guaranteeing the existence and uniqueness of a smooth solution $F(t, s)$.

The stock price is assumed to follow a geometric Brownian motion, meaning its dynamics are driven by a standard Wiener process $W(t)$. To understand why we take a step back and return to the concept of diffusion. A stochastic process X is called a diffusion if its local dynamics can be approximated by a stochastic difference equation of the form:

$$X(t + \Delta t) - X(t) = \mu(t, X(t)) \Delta t + \sigma(t, X(t)) Z(t)$$

where $Z(t)$ is a normally distributed disturbance term, independent of the history of the process up to time t and μ, σ are deterministic functions as described above. We are saying, hence, that over an infinitesimal time step, the X -process is driven by a locally deterministic velocity and a Gaussian

¹⁵ The general definition of partial differential equation, including the notion of classical solution, follows the standard presentation in Evans (2010, Ch. 1). The connection between SDEs and PDEs via the Feynman–Kac formula is discussed in Øksendal (2003, Ch. 8).

disturbance. To model this Gaussian disturbance formally, we introduce the Wiener process. One might attempt to write:

$$\Delta W(t) = v(t), \quad v(t) = dW/dt$$

treating $v(t)$ as the formal time derivative of W . If v were an ordinary, well-defined process, one could in principle solve the resulting equation for X – with $\Delta W(t)$ replacing $Z(t)$ – as a standard ODE along each trajectory. However, it can be shown that a Wiener process is nowhere differentiable so the derivative $v(t)$ cannot be defined in the classical sense. A natural alternative is to define the X -process via integrals rather than derivatives. The system (1) is interpreted as a shorthand version of an integral equation. Unfortunately, a further obstruction arises: the Wiener process has locally unbounded variation, so the integral with respect to dW cannot be defined in the classical Riemann-Stieltjes sense. As long as we insist on giving a precise meaning to the X -process equation for each W -trajectory separately, we are in a hopeless situation. By relaxing this requirement – that is, by defining the integral in a mean-square sense rather than trajectory by trajectory – one can proceed. This leads to the construction of the Ito's integral:

$$\int_0^t g(s) dW(s)$$

We are now in a position to introduce the geometric Brownian motion, which will be one of our fundamental building blocks for option pricing. Its dynamics are governed by the SDE:

$$dX(t) = \alpha X(t) dt + \sigma X(t) dW(t), \quad X(0) = x_0$$

The geometric Brownian motion can be viewed as a linear SDE whose coefficients are proportional to the current state $X(t)$. For small values of σ , sample paths remain close to the deterministic trend; for large values of σ , sample paths remain close to the deterministic trend; for large values of σ , substantial random deviations arise. This assumption is what renders the price process stochastic and gives rise to the need for Itô's calculus. Formally, the stock price satisfies:

$$dS(t) = \alpha(t)S(t) dt + \sigma(t)S(t) dW(t)$$

where $W(t)$ is a standard Brownian motion on a filtered probability space. Without this assumption, the Itô correction term, and hence the entire Black-Scholes framework, would not apply.

To derive the Black-Scholes theorem, the portfolio must satisfy a self-financing condition, which requires that cash is not withdrawn from or injected into the portfolio over time, i.e. assuming a null consumption over $[0, T]$. This condition ensures that the portfolio value evolves purely gains and losses on the held assets, with no exogenous cash flows. As will be shown, it is precisely this property

that allows one to equate the portfolio value to the derivative price at all times and thereby derive the Black-Scholes PDE.

2.3 European Call Options and Contingent Claims

When we first introduced European Call options, a formal definition wasn't necessary for the purpose of explaining market completeness. We now present this financial derivative with a precise theoretical definition, reserving the mathematical formulation for the section that follows.

A European call option with exercise price (or strike price) K and time of maturity (exercise date) T on the underlying asset S is a contract defined by the following clauses:

- *The holder of the option has, at time T , the right to buy one share of the underlying stock at the K from the underwriter of the option*
- *The holder of the option is in no way obliged to buy the underlying stock*
- *The right to buy the underlying stock at the price K can only be exercised at the precise time T*

Considering the perspective of our work, the next step is to translate the European call options into mathematical objects. Since call options are completely determined by the underlying asset S , it is natural to recognize them as a specific instance of contingent claims. We now give the formal definition.

Consider a financial market with vector price process S . A contingent claim with date of maturity (exercise date) T , also called a T -claim, is any stochastic variable $X \in F_T^S$. A contingent claim $X \in F_T^S$ -measurable is called a simple claim if it is of the form

$$X = \Phi(S(T))$$

where $\Phi: \mathbb{R} \rightarrow \mathbb{R}$ is called the contract function.

The word "contingent" is precisely the right one here: the payoff of the claim depends entirely on what happens to the stock price, that is, the payment is contingent on the realised outcome. For a European call option, the contract function takes the form:

$$\Phi(x) = \max(x - K, 0)$$

This encodes exactly the optionality described earlier: the holder exercises only when it is profitable to do so. We now turn to the pricing of the option. We denote the price of a contingent claim X at time t by:

$$\Pi(t; X).$$

For the simplest class of claims – those depending only on $S(T)$ and the time T – we can determine the option price at maturity explicitly. At time T , if the underlying stock price exceeds K , the holder exercises the option, acquires a share at price K and immediately sells it at the market price $S(T)$, realizing a profit of $S(T) - K$. If instead, $S(T) \leq K$, the option is worthless and is left unexercised. This gives

$$\Pi(T) = \max [S(T) - K, 0] \quad (2)$$

which shows the value of the option in T , that is for a general T -claim, the terminal condition is simply

$$\Pi(T; X) = X,$$

and in the particular case of a simple claim depending only on $S(T)$

$$\Pi(T; X) = \Phi(S(T)).$$

We described the value of the option at maturity. However, it is far from obvious what should be the correct price of an option before maturity and this is what we show in the next subsection.

2.4 Pricing the claim before Maturity

The exact price of the option will depend on many factors, for example, the attitude to risk and expectations about future prices. The Black-Scholes theorem takes into account these aspects entirely. Given the market model specified below and the structural assumption that the market is complete and arbitrage-free, we provide a unique, deterministic price for the option at every time $t \in [0, T)$.

2.5 Black-Scholes Model market and absence of arbitrage

The Black-Scholes market model consists of three instruments: a bond, a stock and a contingent claim. We recall the price dynamics of European call options given in formula (2) and we now we look at the bond and the stock.

The price process of the bond B satisfies

$$dB(t) = r(t) B(t)dt, \text{ that is}$$

$$dB(t)/dt = r(t) B(t)$$

Here, the division by dt is fully legitimate, since B is a differentiable function of time in the classical sense. The quantity $r(t)$ is the instantaneous risk-free interest rate: if $r(t)$ is deterministic, $B(t)$ is the

price of a standard bond growing at a known rate, if $r(t)$ is stochastic, $B(t)$ is proportional to the stochastic discount factor. The price process of B has no diffusion term (no Wiener process appears), hence there is no instantaneous randomness in the equation. Over an infinitesimal interval, the increment of B is known with certainty, and the bond price grows smoothly and deterministically. The price process of the stock S satisfies the (1) formula. At time t , the coefficients $\alpha(t, S(t))$ and $\sigma(t, S(t))$ are assumed known (as we underlined in one of the previous sections) but this does not mean that the full increment dS is known. Since $dW(t)$ is a random shock, the realized increment of S over the next infinitesimal interval remains uncertain. What is known at time t is the current value $S(t)$, not the random disturbance immediately following. The stock, therefore has a stochastic rate of return even on the infinitesimal scale.

$$dS(t)/[S(t) dt] = \alpha(t, S(t)) + \sigma(t, S(t)) dW(t)/dt$$

but this expression is strictly heuristic and has no rigorous meaning. The Wiener process W is nowhere differentiable with probability 1 so the ratio dW/dt does not exist as a mathematical object. This is precisely why Itô calculus is required. The correct interpretation of the SDE is that, over an infinitesimal interval dt , the relative return of S decomposes as

$$dS(t) / S(t) = \alpha(t, S(t))dt + \sigma(t, S(t))dW$$

where the stochastic increment dW is of order $(dt)^{1/2}$, not dt , and therefore can never be divided away.

Regarding the absence of arbitrage, the definition has already been established, a portfolio h constitutes an arbitrage if

$$V^h(0) = 0, \quad P(V^h(T) > 0) > 0, \quad P(V^h(T) \geq 0) = 1$$

2.6 Black-Scholes Pricing Formula

The goal of this section is to derive the Black-Scholes pricing formula for a contingent claim. We will do so by constructing a self-financing portfolio that, under the assumption of no arbitrage, forces the price process of the derivative to satisfy a specific partial differential equation (the Black-Scholes PDE). The solution to this equation, with appropriate boundary conditions, gives us the unique no-arbitrage price of the claim. The derivation will proceed in four logical steps.

We begin by studying the dynamics of a self-financing portfolio, as this will produce a technical result, needed for the Black-Scholes derivation. Consider a financial market consisting of several assets, stocks, bonds of different maturities, or financial derivatives, the market value of the portfolio

immediately before rebalancing equals the cost of the new holdings immediately after. In discrete time, this budget constraint reads:

$$S(t) \Delta h(t) = 0$$

where $\Delta h(t)$ represents the change in holdings where we set the consumption rate is zero. Since our goal is to obtain the budget equation in continuous time, we let $\Delta t \rightarrow 0$. However, this limit cannot be performed using classical calculus, because the equation contains Wiener increments — increments that are not differentiable in the classical sense, as already discussed. All stochastic differentials must instead be interpreted in the Itô sense. Therefore, by Itô's calculus and the continuous-time self-financing condition

for a self-financing portfolio $h(t) = (h(B(t)), h(S(t)))$ with no consumption, the portfolio value $V(t) = h(B(t))B(t) + h(S(t))S(t)$ satisfies the stochastic differential equation

$$dV(t) = h(B(t)) dB(t) + h(S(t)) dS(t).$$

In this fundamental self-financing condition in continuous time, the infinitesimal change in portfolio value comes entirely from price changes of the held assets, with no external injection or withdrawal of funds. Having introduced Itô's calculus as a necessary tool, we now take the opportunity to examine it more carefully. Doing so will reveal yet another instance of the central theme of this thesis: mathematics can describe financial reality, but not all mathematics can. Let us consider the price dynamics of a stock

$$dS(t) = \alpha S(t) dt + \sigma S(t) dW(t).$$

The presence of $W(t)$ is what makes this model realistic, since it captures the genuine randomness of financial markets. Yet, this same random term creates a profound mathematical problem, since a Brownian path is nowhere differentiable with probability 1 and the classical calculus, which is built entirely on the assumption of smooth and differentiable paths, breaks down. The ordinary chain rule cannot be applied and a new calculus, built specifically for this noisy and irregular setting, is required. That calculus is Itô's calculus¹⁶. To understand why, recall that in ordinary calculus one approximates the change in a function f over a small step Δx using a Taylor expansion:

$$\Delta f = f'(x) \cdot \Delta x + \frac{1}{2} f''(x) \cdot (\Delta x)^2 + \dots$$

¹⁶ The definition and properties of Itô's calculus, including the Itô formula as the analogue of the classical chain rule, follow Øksendal (2003, Ch. 3) and Karatzas & Shreve (1991, Ch. 3). See also the Brownian Motion notes above,

In the classical setting, as $\Delta x \rightarrow 0$, the term $(\Delta x)^2$ vanishes much faster than Δx , and is safely discarded, leaving the familiar rule $df = f'(x) dx$. For a Brownian motion, this reasoning fails. Since $W(t)$ is a Wiener process whose sample paths are Brownian paths, we have by the independent Gaussian increments property that $E[(\Delta W(t))^2] = \Delta t$, meaning the second-order term is of the same order as Δt , and does not vanish. Discarding it would introduce a systematic error. The core rule of Itô's calculus encodes precisely this observation

$$(dW(t))^2 = dt.$$

All higher-order terms, such as $dt dW(t)$ and $(dt)^2$, remain negligible and can still be discarded, but the second cannot. This observation leads to the Itô's lemma which is the central theorem of the theory serving as the rigorous replacement for the chain rule in a stochastic setting.

Theorem: Itô's Formula

Let $f(t,x)$ be a function of class $C^{1,2}$, and let $X(t)$ be a stochastic process satisfying

$$dX(t) = \mu(t) dt + \sigma(t)dW(t)$$

Then,

$$df(t, X(t)) = (\partial f / \partial t + \mu(t) \partial f / \partial x + \frac{1}{2} \sigma^2(t) \partial^2 f / \partial x^2) dt + \sigma(t) \partial f / \partial x dW(t),$$

where $\partial f / \partial t$ and $\partial f / \partial x$, $\partial^2 f / \partial x^2$ are the partial derivatives of f evaluated at $(t, X(t))$.

The term $\frac{1}{2} \sigma^2 \partial^2 f / \partial x^2 dt$ is known as the Itô correction term. It has no analogue in classical calculus. It exists solely because of the roughness of Brownian motion and it is precisely the term that classical calculus would wrongly discard. As we will see, it plays a decisive role in the derivation of the Black-Scholes equation. It was worth pausing to appreciate what Itô's calculus represents at a deeper level. Its very existence is a testament to the fact that sometimes mathematics does not arrive ready to describe the world, the world itself forces mathematics to grow.

Step 1: Market Model

Consider α , Φ , F , r exogenously given, we recall the market defined in Section 2.6, consisting of a bond and a stock with dynamics:

$$dB(t) = r B(t) dt$$

$$dS(t) = \alpha S(t) dt + \sigma S(t) dW(t)$$

where r , α and σ are assumed constant and deterministic. We introduce a simple contingent claim $X = \Phi(S(T))$ and make two assumptions: the claim is tradable in this market, and its price process $\Pi(t)$ takes the form:

$$\Pi(t; X) = F(t, S(t))$$

That is, the derivative price is a deterministic function of time t and the current price of the underlying asset $S(t)$. For derivation to proceed rigorously, F must be sufficiently smooth:

Definition of F :

Let $U \subset \mathbb{R}^n$ be an open set. A function

$$F: U \rightarrow \mathbb{R}$$

is called smooth if

$$F \in C^\infty(U), \text{ and } U \subseteq \mathbb{R}^2$$

Therefore, all partial derivatives of every order of F with respect to t and S that will be needed, must exist and be continuous on U (there are no jumps, corners, sharp edges). In particular, in our model, we require to differentiate the F function multiple times for S and t . It is sufficient that:

$$F \in C^{1,2}$$

$\partial f/\partial t$, $\partial f/\partial x$, $\partial^2 f/\partial x^2$ exist and are continuous.

Second passage: Creation of a hypothetical self-financing portfolio

Having assumed $\Pi(t)=F(t,S(t))$, we study how $\Pi(t)$ evolves over time by applying Itô's lemma to the F with the dynamics of $S(t)$. This yields:

$$d\Pi(t) = \alpha_\pi(t) \Pi(t) dt + \sigma_\pi(t) \Pi(t) dW(t)$$

where the drift and diffusion coefficients of the derivative are given by:

$$\alpha(\pi(t)) = [F_t + \alpha S F_S + \frac{1}{2} \sigma^2 S^2 F_{SS}] / F$$

$$\sigma_\pi(t) = [\sigma S F_S] / F$$

We now form a portfolio consisting of the underlying stock and the derivative, with portfolio weights $(u(s), u(\pi))$ satisfying $u(s) + u(\pi) = 1$. The value V of this portfolio evolves as:

$$dV = V [u(s)\alpha + u_\pi (\pi\alpha)] dt + V [u(s)\sigma + u_\pi(\pi\sigma)] dW$$

The key observation is that both brackets above are linear in the arguments $u(s)$ and $u(\pi)$. We exploit this by imposing two conditions simultaneously: the portfolio is fully invested:

$$u(s) + u(\pi) = 1$$

and the random dW -term vanishes

$$u(s)\sigma + u(\pi)\sigma(\pi) = 0$$

By eliminating the stochastic component, we have constructed a locally riskless portfolio, whose value evolves purely deterministically:

$$dV = V [u(s)\alpha + u(\pi)\alpha(\pi)] dt$$

Solving the linear system gives:

$$u(s) = \sigma_\pi / [\sigma_\pi - \sigma], \quad u(\pi) = -\sigma / [\sigma_\pi - \sigma]$$

Substituting into the drift, the instantaneous rate of return of the locally riskless portfolio is:

$$k(t) = u(s(t)) \alpha(t) + u(\pi(t)) \alpha(p(t)) = [\alpha(t)\sigma(p(t)) - \alpha(p(t))\sigma(t)] / [\sigma(p(t)) - \sigma(t)]$$

$k(t)$ is the instantaneous rate of return of the locally riskless portfolio, which depends on α , σ , $\alpha(p)$, and $\sigma(p)$, hence it is fully determined by the drift and diffusion coefficients of the stock and the derivative.

Step 3: Imposing The no-arbitrage condition

We have constructed a locally riskless portfolio. In an arbitrage-free market, any locally riskless portfolio must earn exactly the risk-free rate r . If it earned more, a trader could borrow at rate r , invest in the portfolio, and lock in a guaranteed profit with zero net investment – a direct violation of the absence of arbitrage assumption. Therefore:

$$k(t) = r(t)$$

This is a crucial step: the no-arbitrage assumption is what transforms k into r , and it is what set the price of the derivative uniquely. Without this assumption, a continuum of prices could be consistent with the observed market; with it, there is only one.

Step 4: Derivation of the Black-Scholes PDE

We now make the no-arbitrage condition explicit by substituting the expressions for $u(s)$, $u(\pi)$ and $\alpha(\pi)$ obtained in Step 2 into the condition $k(t) = r(t)$. After simplification, all terms reduce to a single equation that F must satisfy:

$$F_t(t, S(t)) + r(t)S(t) F_s(t, S(t)) + \frac{1}{2}\sigma^2(t, S(t))S^2(t)F_{ss}(t, S(t)) - r(t)F(t, S(t)) = 0$$

Writing x in place of $S(t)$, this becomes:

$$F_t(t, x) + rx F_x(t, x) + \frac{1}{2}\sigma^2x^2 F_{xx}(t, x) - rF(t, x) = 0$$

subject to the terminal boundary condition:

$$F(T, s) = \Phi(s) \quad \text{for all } x \geq 0$$

The boundary condition says that at maturity T , the derivative price must equal its contractual payoff, which is simply the definition of a contingent claim. The PDE itself is a deterministic equation in FF F , and it contains a striking feature: the drift α of the stock has completely disappeared. The price of the derivative depends only on the volatility σ and the risk-free rate r , not on the expected

return of the underlying asset. This is one of the first and most celebrated and counterintuitive results in all of financial mathematics.

Theorem:

Consider the market model consisting of a bond $B(t)$ with dynamics $dB(t) = r(t) B(t) dt$ and a stock $S(t)$ with dynamics $dS(t) = \alpha(t) S(t) dt + \sigma(t) S(t) dW(t)$, where $r(t)$, $\alpha(t)$, and $\sigma(t)$ are deterministic functions of time with $\sigma(t) > 0$ for all $t \in [0, T]$. Let $X = \Phi(S(T))$ be a simple contingent claim with maturity T . Assume that the claim's price process takes the form $\Pi(t) = F(t, S(t))$, where $F \in C^{1,2}([0, T] \times \mathbb{R}^+)$. Then, under the no-arbitrage condition, F must satisfy the Black-Scholes PDE:

$$F_t(t, x) + r(t)x F_x(t, x) + \frac{1}{2} \sigma^2(t) x^2 F_{xx}(t, x) - r(t) F(t, x) = 0$$

for all $(t, x) \in [0, T) \times \mathbb{R}^+$, subject to the terminal boundary condition

$$F(T, x) = \Phi(x) \quad \text{for all } x \geq 0.$$

Conversely, if F is a solution to this boundary value problem and the associated replicating portfolio is self-financing, then $\Pi(t) = F(t, S(t))$ is the unique arbitrage-free price of the claim X at time t .

2.7 Solution of the Black-Scholes Equation

The Black-Scholes theorem yields the price process of a European call option in the form of a function $F(t, S)$ satisfying a PDE. There is more than one way to solve this equation; here we proceed via the Feynman-Kac formula, a result due to two mathematicians, Richard Feynman and Mark Kac, both affiliated with Cornell University at the time of its development. The Feynman-Kac theorem establishes a precise correspondence between a broad class of linear parabolic PDEs and conditional expectations of functionals of diffusion processes. It represents the solution to such a PDE as an expected value under the risk-neutral measure.

Martingale measure and Expected Payoff

Before presenting the resolution of our equation, we take a step back to lay out the foundations of the Feynman-Kac formula. This mathematical approach is grounded in the principle of risk neutral valuation, a pricing technique built on the following idea: in an arbitrage-free market, the price of any derivative can be computed as though investors were indifferent to risk. A risk-neutral investor cares about expected payoffs and not about the uncertainty surrounding them. Under this assumption, the price of a derivative today equals the expected value of its future payoff, discounted at the risk-free rate where the expectation is taken under a specific probability measure in which all assets grow, on average, at that same risk-free rate. However, in reality, a risky asset must offer investors a return in excess of the risk-free rate to compensate them for bearing risk and this excess is the risk premium. For pricing purposes, however, we can construct a virtual world in which this premium is set to zero. The asset still fluctuates randomly, but its average rate of growth equals exactly the risk-free rate r .

To achieve this, we replace the real-world probability measure P with the risk-neutral probability measure Q , also called the martingale measure. Under Q , the discounted stock price is a martingale. The purpose of this construction is to replicate the derivative synthetically, so that its price is determined independently of any individual investor's risk preferences. A further essential point is that, as we already established, the market is complete thus we can replicate the derivative which, in an arbitrage-free environment, determines the price as equal to the discounted expectation under risk-neutral probability. We can say that the current stock price equals the discounted expected future price under Martingale (risk-neutral) probability Q . Now, we give the formal definitions¹⁷:

Definition of Martingale:

A discrete time random process $\{X(n); n=0, 1, \dots\}$ is a martingale if

$$E [X(m) / F(n)] = X(n), \quad \text{for all } n \leq m$$

where F_n denotes the information available at time n

Definition of Martingale Measure:

Consider a market model that only exists in time $t=0$ and $t=1$, with N different assets and the price per unit of asset No.1 at time t will be denoted by $S^i(t)$. The randomness in the system is modeled by assuming that a finite sample space $\Omega = \{\omega_1, \dots, \omega_M\}$ and that the probabilities $p(j) = P(\omega_j) > 0$, for $j=1, \dots, M$ are all strictly positive. We say that a probability measure Q defined on Ω is a martingale measure if it satisfies the following conditions

1. *Q is equivalent to P , i.e.*

$$Q \sim P$$

2. *For every $i = 1, \dots, N$, the normalized asset price process*

$$Z^i(t) = S^i(t) / S^1(t)$$

is a martingale under the measure Q .

Solution by Feynman-Kac formula

The Feynman-Kac formula¹⁸ states that the solution to a PDE of a specific structure can be represented as a conditional expectation. More precisely: if a PDE involves a time derivative, a second-order

¹⁷ The definition of martingale process, follows the standard presentation in Karatzas & Shreve (1991, Ch. 1) and Revuz & Yor (1999, Ch. 2). The notion of equivalent martingale measure, its connection to no-arbitrage via the Girsanov theorem, and its role in risk-neutral pricing follow Björk (2009, Ch. 10) and Shreve (2004, Ch. 5). A treatment in the context of general asset pricing theory is given in Duffie (2001, Ch. 6).

¹⁸ The following presentation is taken directly from Øksendal (2003, Ch. 8) and Shreve (2004, Ch. 5). The original formulation of the option pricing problem is due to Black & Scholes (1973). The general statement of the Feynman-Kac theorem in the context of parabolic PDEs can be found in Evans (2010, Ch. 7) and Karatzas & Shreve (1991, Ch. 5).

spatial derivative weighted by the square of a diffusion coefficient, a first-order spatial derivative weighted by a drift coefficient and a discounting term, then its solution at any point (t,x) equals the expected value of the terminal condition, evaluated along a diffusion process starting at x at time t , discounted at the appropriate rate. This is a two-directional bridge: every such PDE has a probabilistic representation as a conditional expectation, and every such conditional expectation satisfies a corresponding PDE. In the Black-Scholes model, the underlying asset price $S(t)$ follows geometric Brownian motion:

$$dS(t) = \mu S(t) dt + \sigma S_t dW(t)$$

The derivative price $V(t,S)$ satisfies the Black-Scholes PDE on the domain $t \in [0,T]$, $S > 0$:

$$\partial V / \partial t + \frac{1}{2} \sigma^2 S^2 \partial^2 V / \partial S^2 + r S \partial V / \partial S - r V = 0$$

with terminal condition $V(T, S) = \Phi(S)$, where Φ is the contractually specified payoff. For a European call option with strike K :

$$\Phi(S) = \max(S - K, 0)$$

The Feynman-Kac theorem, in its general form, applies to PDEs of the structure:

$$\partial u / \partial t + \mu(t,x) \partial u / \partial x + \frac{1}{2} \sigma(t,x)^2 \partial^2 u / \partial x^2 - r(t,x) u = 0, \quad u(T, x) = \Phi(x)$$

Under appropriate regularity conditions, the solution $u(t,x)$ admits the stochastic representation:

$$u(t, x) = E \left[e^{-\int_t^T r(s, X(s)) ds} \Phi(X(T)) \mid X(t) = x \right]$$

where $X(s)$ satisfies $dX(s) = \mu(s, X(s)) ds + \sigma(s, X(s)) dW(s)$ and the expectation is taken under the probability measure associated with that process. Before applying Feynman-Kac, a preliminary step is generally required: the theorem demands that the risk-free rate r appear both as the drift of the underlying process and as the discounting rate. In general, this requires a change of probability measure from the real-world measure P to the risk-neutral measure Q , which is made possible by Girsanov's theorem. In our case, however, this step has already been carried out implicitly. By constructing an instantaneously riskless portfolio and equating its return to r , the no-arbitrage condition caused the true drift μ to drop out of the PDE entirely. What remains has r as the drift coefficient of the underlying process, regardless of the asset's true expected return. The volatility σ , however, is unchanged. The Black-Scholes PDE is therefore already expressed under the risk-neutral measure Q , and Feynman-Kac can be applied directly.

Matching the Black-Scholes PDE against the general Feynman-Kac form, we can identify $\mu(t,S)=rS$, $\sigma(t,S)=\sigma S$ and $r(t,S)=r$. The corresponding diffusion is geometric Brownian motion under Q , and the theorem immediately yields:

$$V(t, S) = e^{-r(T-t)} E^Q[\Phi(S(T)) | S(t) = S]$$

This is the fundamental pricing principle: the price of the derivative equals the risk-neutral expected present value of its future payoff. The Feynman-Kac formula is precisely what justifies this as the solution to the PDE. Under Q , geometric Brownian motion has the explicit solution:

$$S(T) = S \exp \left((r - \frac{1}{2}\sigma^2)(T-t) + \sigma(\tilde{W}(T) - \tilde{W}_t) \right)$$

so that $\log(S(T)/S(t))$ is normally distributed under Q with mean $(r - \frac{1}{2}\sigma^2)(T-t)$ and variance $\sigma^2(T-t)$. The asset price at any future time therefore follows a lognormal distribution whose parameters are entirely determined by the current price S , the risk-free rate r , the volatility σ , and the length of the time interval $T-t$. This makes the conditional expectation analytically tractable. For a European call with payoff $\Phi(S(T)) = \max(S(T) - K, 0)$, the expectation becomes an integral against the lognormal density f_Q of $S(T)$ under Q :

$$V(t, S) = e^{-r(T-t)} \int_0^\infty \max(s - K, 0) f_Q(s) ds$$

By completing the square in the exponent of the lognormal density, this integral separates into two Gaussian integrals, each expressible in terms of the cumulative distribution function N of the standard normal. These two arguments are conventionally denoted as d_1 and d_2 :

$$d_1 = \left[\ln(S/K) + (r + \frac{1}{2}\sigma^2)(T-t) \right] / (\sigma(T-t)^{1/2}), \quad d_2 = d_1 - \sigma(T-t)^{1/2}$$

The integral evaluates in closed form to the celebrated Black-Scholes formula:

$$V(t, S) = S \cdot N(d_1) - K e^{-r(T-t)} N(d_2)$$

The power of the Feynman-Kac approach lies in the equivalence it establishes between two seemingly different mathematical objects. The Black-Scholes PDE is a deterministic equation: it makes no explicit reference to randomness. Yet its solution is naturally expressed as a stochastic expectation. The PDE encodes, through its structure, all the statistical information about the future evolution of the asset price under Q ; Feynman-Kac decodes this information and translates it back into probabilistic terms. It is, in this sense, the deepest illustration of the central theme of this thesis: mathematics and finance, each incomplete on their own, become fully coherent only when brought together.

2.8 Limits of the Black-Scholes model

The Black-Scholes model possesses mathematical elegance and is widely adopted as a benchmark in the derivatives industry. However, it rests on a set of simplifying assumptions that, in real financial

markets, are systematically and persistently violated. These structural limitations are not merely theoretical technicalities: they have direct and measurable consequences on the reliability of the prices produced by the formula. The principal limitations are the following.

The model assumes that the volatility σ of the underlying asset is a known and constant parameter throughout the entire life of the option. In practice, volatility is a fundamentally stochastic quantity varying over time and fluctuating in response to macroeconomic announcements, shifts in market sentiment, changes in trading volume and asset-specific information flows.

To see why this matter, recall how σ enters the pricing formula:

$$V(t, S) = S \cdot N(d_1) - K e^{-r(T-t)} N(d_2)$$

where

$$d_1 = [\ln(S/K) + (r + \frac{1}{2}\sigma^2)(T - t)] / (\sigma\sqrt{T - t}), \quad d_2 = d_1 - \sigma\sqrt{T - t}$$

The parameter σ is not confined to a single term, it appears in both d_1 and d_2 , entering through the numerator, the denominator and the correction term that links the two. Through $N(d_1)$ and $N(d_2)$, it shapes the entire distribution of future asset prices. In particular, a higher σ spreads that distribution wider, making extreme outcomes more likely and therefore increasing the option price. Crucially, the model uses one single σ for every option written on the same underlying asset, regardless of its strike K or maturity T . The most direct way to test whether this is a reasonable description of reality is through the concept of implied volatility. Since the Black-Scholes price is a strictly increasing function of σ , the formula can be inverted: given an observed market price for an option with a specific strike K and maturity T , there exists a unique value $\sigma_{\text{imp}}(K, T)$ that makes the formula match that price exactly. If the model were correct, this number should be the same for all options on the same underlying since they are all governed by the same constant σ , entering identically in d_1 and d_2 regardless of which contract is being priced. This prediction is systematically violated in every major derivatives market. When σ_{imp} is computed across options with the same maturity but different strikes and plotted as a function of K , the resulting curve is never flat. For equity indices, it is typically downward sloping: implied volatility is higher for options with strikes below the current asset price and lower for options struck near or above it. For these indices, the curve has a shape known as the volatility skew. For foreign exchange options, the curve tends to be U-shaped and symmetric, with higher implied volatilities at both ends of the strike range relative to the at-the-money level, a shape known as the volatility smile. Both patterns reflect the same underlying problem: the market prices different options as though they were governed by different levels of volatility, which is only consistent with a σ is not truly constant. The assumption is therefore not merely imperfect, it is structurally incompatible with what markets produce. Despite this, Black-Scholes remains widely used in practice, precisely because of the invertibility argument described above. Even if the model

is not literally correct, it still provides a direct and universally understood link between an option price and a volatility figure. Traders can therefore use σ_{imp} as a convenient way to quote and compare option prices across the market, without committing to any belief that volatility is actually constant. Similarly, the Black-Scholes model treats the risk-free interest rate r as a known deterministic constant, when in reality, interest rates evolve stochastically in response to central bank policy decisions, inflation expectations and macroeconomic conditions. To understand this, we observe where r appears in the formula.

$$V(t, S) = S \cdot N(d_1) - K e^{-r(T-t)} N(d_2)$$

The term $e^{-r(T-t)}$ is the discount factor: it brings the strike price K back to today's value. For short-dated options (where $T-t$ is small) this exponent is a tiny number and the discounting factor remains very close to 1 regardless of small movements in r . For example, a shift of 0.5% in the interest rate translates into a negligible change in the option price, making the constant rate assumption largely harmless. In those cases, the benefits of working with a simpler model outweigh the pricing error it introduces. This is no longer true when we apply the model to long-dated derivatives. When the difference between T and t is large, let's say, 10 years, the same 0.5% shift in r , multiplied over a long horizon, produces a meaningful change in the discounting factor, which directly distorts the theoretical price in a way that can no longer be ignored. The stochastic nature of interest rates can therefore have a significant effect on both pricing and hedging for long-maturity instruments. For this reason, dedicated term-structure models, such as those of Vasicek or Cox, Ingersoll and Ross, are specifically designed to handle this source of risk.

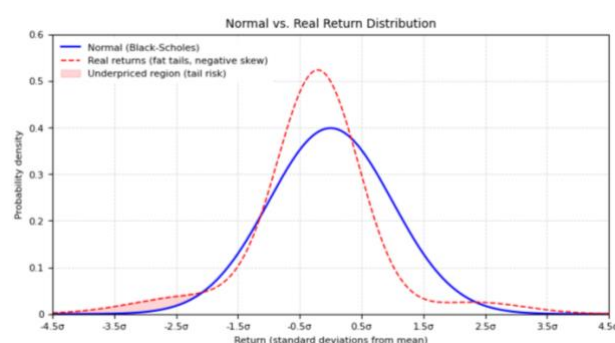
The Black-Scholes derivation assumes a complete and frictionless market which means that the underlying asset can be traded continuously, in arbitrary quantities, without transaction costs, taxes or bid-ask spreads. To understand why this assumption is necessary, we recall what the model requires. The theoretical justification of the Black-Scholes formula rests on the construction of a replicating portfolio, which holds a continuously adjusted position of $N(d_1)$ units of the underlying asset and a corresponding bond position, calibrated so that the portfolio exactly mimics the option payoff at all times. Maintaining this replication perfectly aligned with the option, requires rebalancing the portfolio at every instant as S and t change. In a frictionless market, this process can be carried out without any costs and any constraints on the size or frequency of trades. In reality, transaction costs are unavoidable, market liquidity is finite hence the continuous rebalancing is operationally infeasible. Every adjustment to the hedge position carries a cost, and the more frequently it is executed, the more those costs accumulate. Despite this, Black-Scholes remains a useful framework in practice. Rather than rebalancing continuously, traders use the quantity $N(d_1)$, known as the delta

of the option, as a guide for how much of the underlying to hold at any given moment, accepting that the hedge will be imperfect and managing transaction costs by rebalancing at discrete intervals.

The last assumption worth noting of the the Black-Scholes model is that the underlying asset follows a geometric Brownian motion, which implies that the logarithm of the asset price is normally distributed under the risk-neutral measure:

$$\ln(S(T)/S(t)) \sim N((r - \frac{1}{2}\sigma^2)(T-t), \sigma^2(T-t))$$

This is not merely a background technical condition, it is the foundation on which the entire formula rests. To see this concretely, recall that $N(d_2)$ in the pricing formula represents the risk-neutral probability that the option expires in the money, that is, that $S(T)$ ends up above the strike K . This probability is computed directly from the lognormal distribution above: the term $\ln(S/K)$ in the numerator of d_2 measures how far the current price is from the strike in log-space, and the result is standardized using the assumed normal distribution. If the true distribution of returns were different, $N(d_2)$ would yield the wrong probability, and the entire price V would be distorted accordingly. This is precisely what the data shows. In financial markets., empirical return distributions exhibit fat tails and, for equity indices, negative skewness relative to the normal benchmark. Extreme price movements, both upward and downward, occur with a frequency that the normal distribution assigns negligible probability to. Since $N(d_2)$ is built on the normality assumption, it systematically underestimates the likelihood of large moves ending up deep in or out of the money. As a direct consequence, the Black-Scholes model underprices options at the extremes of the strike range, precisely where tail risk is most concentrated and where the gap between the assumed and the true distribution is most consequential. This mispricing is not random noise: it is the same structural distortion that, when observed through implied volatilities across strikes, produces the skew and smile patterns described in the previous section. The lognormality assumption and the volatility surface anomalies are therefore two manifestations of the same underlying deficiency of the model.



2.9 Implications of the limits and possible extensions

The limitations described above are not without consequence for the prices produced by the Black-Scholes formula. A model that does not correctly describe the statistical properties of the underlying asset cannot be expected to produce option prices that consistently agree with those observed in the market. The resulting discrepancy between model price and market price manifests with particular severity under specific conditions. The most direct consequence follows from the distributional failure discussed in the previous section. Since Black-Scholes assigns too little probability mass to the left tail of the return distribution (remember that $N(d_2)$ is computed entirely under the lognormal assumption) it systematically underprices options that pay off in precisely those scenarios. Deep out-of-the-money put options, which provide protection against large downward moves, are consistently cheaper under the model than at their traded market levels, because the model underestimates the probability of the underlying reaching their strike. This mispricing is not random: it is structural and directional, and it widens exactly when it is most consequential. During periods of high volatility, for example those arising during financial crises or sharp market corrections, the Gaussian assumption breaks down most dramatically; tail events occur with a frequency far exceeding what the normal distribution assigns to them, and the gap between Black-Scholes prices and observed market prices is at its largest. This is also the mechanism behind the implied volatility skew described earlier. Since no single constant σ can simultaneously explain the market prices of options across all strikes, inverting the formula on observed prices yields a different implied volatility for each contract, higher for low strikes where the model underprices most severely, lower for strikes near or above the current asset level. The skew is therefore not an independent phenomenon but a direct reflection of the model's distributional mispricing, expressed in the language of implied volatility. These limitations have motivated a rich line of theoretical research aimed at extending the Black-Scholes framework while preserving its absence of arbitrage foundations. Three families of models are particularly prominent. Stochastic volatility models, of which the most widely adopted is the Heston model (1993), replace the constant σ with a second stochastic process for the instantaneous variance. This allows volatility to evolve randomly over time and generating a skew naturally through the correlation between volatility and price movements. Local volatility models, introduced independently by Dupire (1994) and Derman and Kani (1994), take a complementary approach. Rather than adding a new source of randomness, they calibrate a deterministic function $\sigma(t, S)$ that reproduces the entire observed market surface exactly by construction. The surface is therefore an input to the model rather than an output. Finally, jump-diffusion models, following Merton (1976), address the heavy tail problem most directly by augmenting the continuous diffusion with a jump component, allowing the asset price to move discontinuously and immediately producing return distributions with fatter tails

than any purely continuous model can generate¹⁹. Each family corrects a specific deficiency of the original framework, and in practice elements of different approaches are often combined. What all of them share, however, is the recognition that Black-Scholes model remains the indispensable reference point against which every more sophisticated model is measured. This is a testimony to the depth and durability of the mathematical structure that this thesis has aimed to illuminate.

¹⁹ If you wish to develop these topics further, a more in-depth treatment can be found in the following texts. The local volatility approach, due independently to Dupire (1994) and Derman & Kani (1994), is discussed in Björk (2009, Ch. 20) and Shreve (2004, Ch. 7). The stochastic volatility approach, and in particular the Heston model, is treated in depth in Gatheral (2006), *The Volatility Surface: A Practitioner's Guide*, Wiley, Hoboken. For the broader question of heavy tails, non-Gaussian distributions, and the limits of models built on Brownian motion, the reader is referred to Taleb (2007) and Mandelbrot & Hudson (2004).

3. Black-Scholes in Python

This is the final chapter of the thesis and it is devoted to a concrete, real-world application of the Black-Scholes pricing model, developed in the second chapter. The objective is to demonstrate that the theoretical construction presented there is not merely an abstract mathematical exercise, but rather a rigorous and operationally relevant instrument employed daily in financial markets. To this end, the model is implemented computationally using the Python programming language within the Google Colaboratory environment and all input parameters are sourced directly from the Bloomberg Terminal.

3.1 Bloomberg: the inputs

The chapter is structured as follows. First, the five input parameters required by the Black-Scholes formula are identified and their financial interpretation is discussed. Second, the methodology adopted to extract each parameter from the Bloomberg Terminal is described in detail. Third, the Python implementation of the model is presented, with each computational step linked explicitly to the corresponding mathematical expression derived in Chapter 2. The chapter concludes with the numerical result for the price of the selected European call option. The Black-Scholes formula for the price of a European call option, as derived in the preceding chapter, is:

$$V(t, S) = S \cdot N(d_1) - K e^{-r(T-t)} N(d_2)$$

where:

$$d_1 = [\ln(S/K) + (r + \frac{1}{2}\sigma^2)(T - t)] / (\sigma\sqrt{T - t}), \quad d_2 = d_1 - \sigma\sqrt{T - t}$$

The formula requires the specification of five parameters: the current price of the underlying asset (S), the strike price of the option (K), the time remaining to maturity expressed in years (T), the continuously compounded risk-free interest rate (r) and the volatility of the underlying asset (σ). Each of these quantities admits a precise financial interpretation and must be carefully sourced from reliable market data.

3.2 Data Extraction

All market data used in this application were retrieved from the Bloomberg Terminal on 23 April 2026. The Bloomberg Terminal is one of the most comprehensive and institutionally trusted sources of real-time and historical financial data and its adoption in this context here ensures the reliability and reproducibility of the results presented. The underlying asset selected for this application is the Euro Stoxx 50 Index (Bloomberg ticker: SX5E), which tracks the performance of the fifty largest blue-chip companies in the Eurozone. This choice is motivated by a fundamental theoretical requirement: the Black-Scholes model, as presented in this thesis, prices European call options, which may only be exercised at maturity. Options written on the Euro Stoxx 50 and traded on Eurex are European-style by construction, making this index the appropriate underlying asset for the purposes of this application. The five parameters were extracted as follows. . The value of S was obtained by navigating to the SX5E equity page on Bloomberg and retrieving the PX_LAST field, which returns the most recent transaction price. The complete daily price history over the preceding year was exported and the most recent observation, corresponding to 23 April 2026, was retained as the input value. The strike price (K) and time to maturity (T) were sourced from the Bloomberg Option Monitor (OMON function), which displays the full option chain for a given underlying asset. It was selected from the available expiry dates and strikes, the option with a strike of 5,900 and an expiration date of 19 June 2026, corresponding to 57 calendar days from the date of data extraction. This choice was motivated by the proximity of the strike to the prevailing index level, placing the option in a near at-the-money configuration, which is the most analytically instructive case for the Black-Scholes framework. The implied volatility (σ) was read directly from the IVM column of the OMON screen, which reports the implied volatility mid-quote for each listed contract. For the selected option, this value was recorded as 18.75%, which was subsequently converted to decimal form (0.1875) for use in the formula. The risk-free interest rate (r) was inferred from the rate parameter displayed in the OMON screen header for the selected expiry group, which Bloomberg reports as R 2.40. This corresponds to an annualised continuously compounded risk-free rate of 2.40%, consistent with short-term Eurozone rates prevailing at the time of data collection. The five input parameters are summarised in the table below:

Parameters	Bloomberg values
S	5857.77
K	5900
T	57/365
σ	0.1875

r	0.0240
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Personal paper based on the Bloomberg data collected

3.3 Python Implementation

The computational implementation of the Black-Scholes model was carried out in Python using the Google Colaboratory platform, a cloud-based interactive notebook environment that requires no local software installation. The code is structured into a sequence of logical steps, each of which corresponds to a distinct phase of the pricing procedure.

First step

The first step consists of importing the necessary computational libraries: NumPy, which provides numerical operations including logarithms and square roots, SciPy, from which the cumulative standard normal distribution function $N(\cdot)$ is obtained, Matplotlib, used for graphical output and Pandas, employed for reading and processing the Excel files exported from Bloomberg. Finally, the Re module library provides regular expression functionality, used here to parse the Bloomberg option chain header row in order to extract the time to maturity and the risk-free rate directly from the file.

Second step

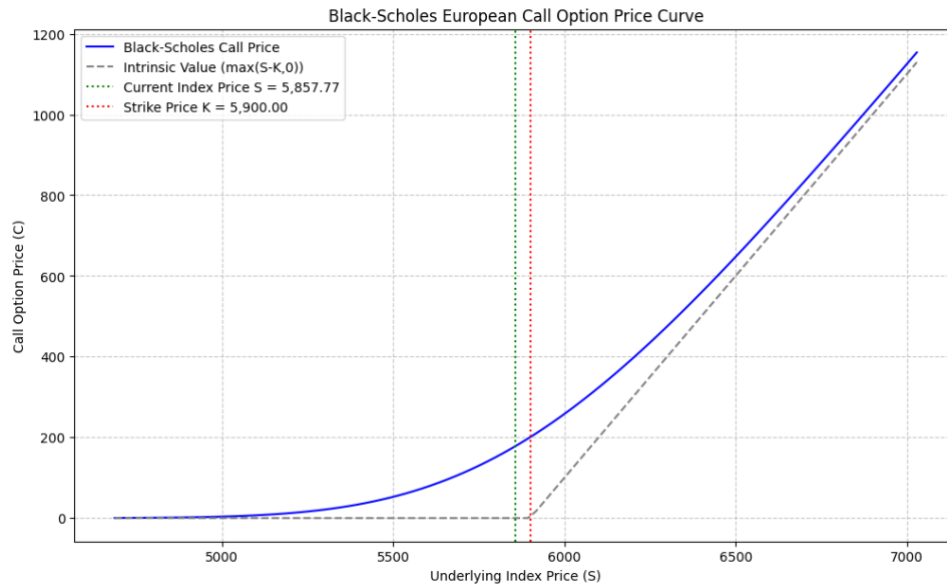
The second step involves loading the Bloomberg data. Both files, containing the information about stock and option chosen, are uploaded interactively within the Colaboratory environment and identified dynamically by filename so that no hardcoded path is required. The stock price file is read into a Pandas DataFrame and the most recent price observation is extracted from the first row, corresponding to the date of data collection. The option chain file requires more careful handling. From it, the code extracts the number of calendar days to maturity, which is divided by 365 to obtain T , and the annualized risk-free rate R , which is divided by 100 to convert it to decimal form. Then, the code continues scanning within the same expiry block until it finds the row corresponding to strike 5,900, from which the strike price K and the implied volatility σ are read. The implied volatility, reported by Bloomberg in percentage form in the IVM column, is divided by 100 as required by the formula.

Third step

The third and central step is the definition of the Black-Scholes pricing function. The function receives the five parameters as arguments and computes, in sequence: the two auxiliary quantities ($d1$ and $d2$) defined in the previous chapter, and finally the call option price C by evaluating the full Black-Scholes expression. Each line of code maps directly to the corresponding term in the analytical formula.

Fourth step

The fourth step calls the function with the five Bloomberg inputs and prints the result in a structured format, displaying all input parameters alongside the computed option price. To better understand how the option price changes with variations in the underlying asset's price, we will plot the Black-Scholes call price curve. This visualization will include the current index price (S) and the strike price (K) for comparison, as well as the intrinsic value of the option.



Personal paper on the Black-Scholes Call Price Curve based on the Bloomberg data collected

Conclusion

I would like to finish this thesis by quoting a famous phrase said by Warren Buffett: “*Risk comes from not knowing what you're doing*”²⁰. The remark carries particular weight precisely because it comes from an investor who built his career not on volatility and quantitative indicators, but rather on the careful assessment of the intrinsic value behind business realities. This suggest that what he says is a truth transcending any single methodology. Risk is intrinsic to every investment that deserves the name, this is the nature of finance. It cannot be eliminated; however, it can be measured. This is exactly the transformative contribution of quantitative finance: it becomes possible not merely to face uncertainty, but to calculate it. We may not be able to remove the risk from the equation, but we can know exactly how much of it we are taking. And when we do, we know what we are doing.

²⁰ Warren Buffett, address to Columbia University Business School, 1993.

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